

Abhishek Kumar

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SUMMARY

Backend Software Engineer with 3 years of experience building **scalable distributed systems** and **AI-powered** applications using **Python, FastAPI, PostgreSQL, Kafka, and async processing**. Experienced in designing high-throughput **APIs**, **real-time systems**, **workflow orchestration**, **RAG pipelines**, and **agentic AI systems**.

WORK EXPERIENCE

Limitless Neurons, Hyderabad | *Software Engineer*

May 2025 - Present

- **Shapes Chat** | *FastAPI, SQLite, Langchain, Langgraph*
 - * **Architected and built a RAG system** enabling users to chat with **real-time RSS feeds**, integrating multiple **LLM providers via LiteLLM**, reducing vendor lock-in and optimizing query costs.
 - * Engineered complex **agentic workflows using LangGraph** to perform autonomous multi-step analysis (**conflict detection, topic clustering**) on aggregated news streams, delivering deep insights beyond simple retrieval.
 - * Implemented a local-first **Vector Search architecture using ChromaDB**, optimizing retrieval precision with hybrid search strategies (**semantic + keyword**) and dynamic context window management.
 - * **Developed a streaming response** infrastructure using **Server-Sent Events (SSE)**, enabling real-time token streaming for **AI responses**, significantly improving perceived user latency.
- **Autogrid** | *FastAPI, SQLite, React, Electron*
 - * **Architected an async execution engine** using Python **asyncio** to orchestrate **100+ concurrent AI/API** tasks across **DAGs** with granular rate limiting, while building a local-first **desktop app** for cross-platform.
 - * **Designed a type-safe Plugin System** with Protocols and Pydantic, standardizing config validation and lowering integration effort by 40% for services like **OpenAI and Clearbit**.

Cogoport, Gurgaon | *Software Development Engineer - 1*

September 2023 - April 2025

- **Business Finance** | *Kotlin, Micronaut, Postgresql, RabbitMQ, FastAPI*
 - * **Developed APIs** using optimized **SQL** queries for the **Centralized Finance system**, supporting payment processing for **100+ daily transactions** and accurate financial reporting.
 - * **Engineered IRN generation** for real-time **e-invoice declaration in China** utilizing the **Aisino platform**, achieving **100%** compliance with tax regulations and slashing **manual intervention by 85%**.
- **Job Card** | *Python, FastAPI, Postgresql, Tortoise-ORM, Apache Kafka, Websockets*
 - * **Leveraged Apache Kafka for real-time data streaming** and designed a **job allocation logic** optimized for operator expertise, stickiness and organizational requirements.
 - * **Implemented WebSockets for real-time communication** between client and server, reduced latency by **90%**.
- **Shipment Execution** | *Python, FastAPI, Postgresql, Peewee*
 - * **Contributed to end-to-end migration of Shipment business logic**; engineered **invoicing APIs**, reducing API runtime by **70%**, and leveraged **Redis, Celery, and Flower** for asynchronous task processing.
 - * **Designed an end-to-end shipment cancellation workflow** coordinating service and downstream state transitions while maintaining transactional consistency.

EDUCATION

Indian Institute of Technology, Delhi

2019 - 2023

Bachelor of Technology in Chemical Engineering

CGPA: 8.386/10

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, JavaScript, Kotlin, SQL

Backend & Frameworks: FastAPI, Micronaut, Spring Boot, Node.js, React

AI/ML: LangChain, LangGraph, RAG, Vector Search, ChromaDB

Databases & ORM: PostgreSQL, SQLite, SQLAlchemy, Peewee, Tortoise

Messaging & Distributed Systems: Apache Kafka, RabbitMQ, Websockets, Redis, Celery

Software Engineering: REST APIs, Microservices, System Design, Design Patterns, Scalability, Reliability

ACHIEVEMENTS

- **Competitive Programming:** LeetCode Knight - **1990 rating**, Solved **~650** problems; Codeforces Specialist - **1547 rating**, Solved **~600** problems.